What is Sybil Attack?

Introduction

A Sybil attack takes place when a person makes many fake accounts to deceive a system. Those fake accounts seem like real ones, all of which are controlled by one person. Purpose is to control and overcome the system, or to destroy it. It is a cunning trick that can create big problems online, particularly in the networks or systems where the trust between the users is of great importance.

Hence, this problem can be resolved through a multiple-approach. Encryption protects private and confidential data from unlawful use and keeps it away from unexpected persons. On the other hand, social network analysis lets administrators identify user behavioral trends, allowing them to distinguish between genuine participants and imposters.

However, it is crucial to note that not all false accounts are malicious some of them might be created for sun only. In this case, defense mechanisms might not work as expected, and this, might divert the attention away from the actual issues.

History of Sybil attack

The name "Sybil crime" was inspired from the novel "Sybil", which was about a woman name. She suffered from dissociative identity disorder. Person named Douceur presented the ideas in his speech called "The Sybil Attack." He was from research team of Microsoft and was very smart.

Few examples of Sybil attacks in different contexts:

- The fake accounts on Facebook that were allegedly operated by the Russians during the U. S. elections.
- Spoofing attacks are a bad phenomenon happened to the Tor network, where bad actors can create multiple fake ID to desire the other Tor nodes.
- When an entity gains control over more than a half of the blockchain network's hash rate (which functions as a voting mechanism within the network), such an attack is called the Sybil attack.
- Another example is, fake reviews on e-commerce platforms like Amazon where a single person creates multiple fake accounts to manipulate product ratings. This happens in very large number.

However, this scenario can be seen by the location of disagreements, which could vary from social media platforms to secret networks, block chain systems, and e-commerce systems. Since it can be hard to differentiate between fake and genuine identities in a decentralized or even fully decentralized system, Sybil attacks attack categories provide big challenges. The key point is that, because of mutual trust, both cryptographic methods and social network-based trust have been proposed. They are not entirely trustworthy, though, because they have consequences.

Types of Sybil attacks

It depends how the attack is performed and what is the target of the attackers. Here are some common types or possible types of Sybil attacks:

- **Direct Sybil Attack:** This kind of attack is one in which the attacker creates multiple identities or nodes right inside the target system or network. The attacker actually tries to have a disproportionate volume of influences through a significant number of pseudo identities.
- **Indirect Sybil Attack:** A Sybil attack, which in the direct form is when the attacker creates identities or nodes outside the target system or network and then tries to introduce them into the system, can be done in the indirect form, or when the attacker creates identities or nodes inside the system but still tries to introduce them. This kind of attack is usually more difficult to find because the identities will look real at first.
- **Simultaneous Sybil Attack:** This attack is unique in that its attacker creates a large number of identities at the same time and uses them in a coordinated attack on the target system or network.
- **Sequential Sybil Attack:** A smart attacker could slowly but surely create lines using identities during the Silent Attack to increase their power within the system or network.
- **Sybil Attack with Identity Theft:** Here, the attacker does not create new fake identities but takes control over the original identities or nodes within the system, thus it makes the whole network vulnerable.
- **Sybil Attack with Colluding Identities:** In these situations, hackers construct multiple identities which operate jointly or independently to achieve a goal, such as seizing power over the majority or affecting the decision-making process.
- **Targeted Sybil Attack:** In this type of Sybil attack, the attacker looks for a vulnerable place in the organization like spreading attack just on a specific part of the system or division. Such an attack is harder to be detected and to be remedied.
- **Pseudo Sybil Attack:** The attacker who need not necessarily create numerous newsy fake accounts exploits weaknesses in identity management or authentication, resulting to the so-called pseudo Sybil attack. After that, it is frequently challenging to identify and decrease the likelihood that these attacks will be successful.

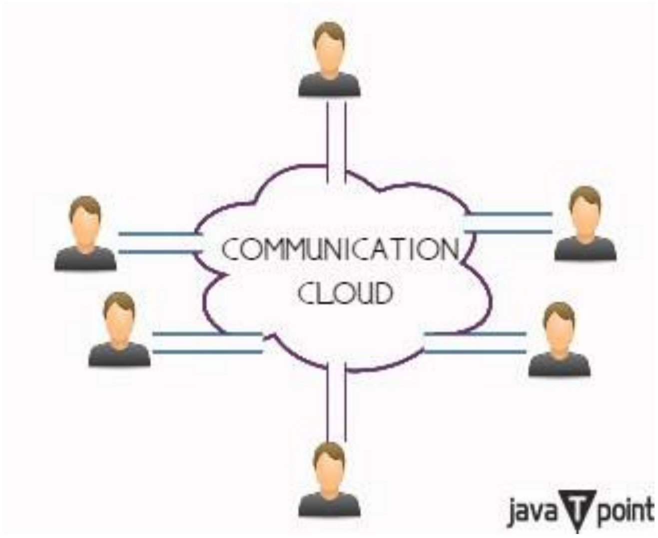
There are various types of Sybil attack as of networks or decentralized systems based on how it is executed. Here we read few examples of the attacks that can happen in various systems and networks. Knowing the different categories of attacks can enable you to protect yourself from various attacks. People should follow all possible safety measure to avoid these attacks, nowadays it is near to impossible to protect yourself fully but safety measures can prevent up to some extent.

Formal model of Sybil attack

The model, in turn, assumes that the world consists of two elements:

- **Correct entities (c):** These participants endorse consistency and the protocols of network, and they can be put to the test through validity check.

- **Faulty entities (f):** These entities display the behavior that is not following the protocols and rules of the network and is not rational.



- **Communication Cloud:** This serves as an example of a general communication medium that serves as both an official tool and a voice for the people. It is frequently employed as a conceptual model of the network backbone architecture.
- **Pipe:** Each component is linked to the communication cloud through the pipe, which enables the flow of messages between the component and the network.

According to this model, the total number of entities (E) is formed up of the faulty/faulty elements (f) and the proper entities (c). The threat of a Sybil attack can be seen by the existence of corrupted entities, as the attacker can create more fake nodes for taking control of the network and disrupt fair governance.

Are Sybil attacks resistant on all Blockchains?

Blockchain are known to be securing than any other platform but attackers always find a way to get into even safest place so we can say that blockchain can also face Sybil attack. It has shown to be extremely difficult to attack Bitcoin using Sybil or a 51% mining attack. Nevertheless, other altcoins that rely on a lower hash rate could be more prone to manipulation and 51% assaults that result in double-spends. Several altcoins, such as Ethereum Classic (ETC), Bitcoin Gold (BTG), Vertcoin (VTC), and Verge (XVG), have seen one or more 51% attacks.

Parallel "Sybil nodes" are set up by the attacker to vote more than genuine votes, which are suppressed or made to wait until the next round of voting, cause communication problems, damage people's reputations, or amass more resources than they ought to.

It gets its name from the fact that it resembles when someone has a soft spot for particular facets of their personality. Sybil attacks are virtually unsolvable since it is nearly hard to distinguish between authentic and fraudulent users in a decentralized system. By disguising as many servers as possible, the goal is to prevent hostile actors from exerting undue influence.

Conclusion

Sybil attacks with lots of fake identities are dangerous, so we need good ways to catch them to keep decentralized networks safe and trusted.